



Data Warehouse Project

General Guidelines:

- The team consists of 5-6 students
- 2 parts to implement in the project:
 - Using Alteryx
 - Using SQL SERVER Integration Services (SSIS)
- Every team member will be asked in the project discussion

Project Description

Main topics: Each team should:

- Extract data from Dataset
- Make Transformation
- Load data in destination
- Construct Star schema
- Make analysis on star schema using Power BI
- Write a documentation

Detailed Requirements:

- Each team should get a dataset (source) to work on from the Internet.
- Each team will be assigned a category to get a dataset from.
- The dataset could be in any format (excel, flat file etc.)
- Use **Alteryx** to **split dataset records into half** and load them in two different destinations. (do not use total number of records in your solution)
 - First and second halves destinations will be in excel format
- For the first half you will use **Alteryx** and for the second half you will use **SSIS**
- **Each team should do the same transformation on the two halves but each on its tool (Alteryx - SSIS)**
- Transformations to use:
 - **1**: filter records on specific condition (custom condition)

- **2:** choose any string column and uppercase the first character only
- **Third:** split any column into many columns (Ex: date)
- **Fourth:** replace any white spaces with underscore for any string column
- Depending on first point **(1)** in transformation, each half will be divided again in two parts.
 - First Half (part 1 and part 2)
 - Second half (part 1 and part 2)
- You will have 2 destinations.
 - Load part 1 from first and second halves in destination 1.
 - Load part 2 from first and second halves in destination 2.
- The two destinations will be in SQL Server.
- Design **Star schema with 2 dimensions** and load data from any destination into it.
- Use Power BI to generate charts from the star schema (at least 3 charts) and you **must** include and apply some of the **DAX** functions that are needed or suitable for your data.
- Make documentation which includes:
 - Short paragraph describing dataset and its columns
 - Which components used in the 2 tools (Alteryx and SSIS)
 - Screenshots from each component used
 - What decisions should be taken depending on charts results

Best Wishes